

Dr. Lorenzo Mansi

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Education

Imperial College London

Master of Science in Statistics

Sep 2026 – Sep 2027

London – UK

- **Relevant Coursework:** Computational Statistics and Machine Learning, Statistical Inference, Probability, Deep Learning, Non-parametric Statistics, Probabilistic Generative Models, Statistical Finance, Stochastic Processes, Time Series Analysis
- **Societies:** member of the Algorithmic Trading Society.

University of Hamburg

Doctor of Philosophy in Theoretical and Mathematical Physics

Oct 2022 – Jul 2025

Hamburg – DE

- Graduated with 1.0/1.0 *Magna cum laude*.
- Research focus on Representation Theory and Algebraic Geometry techniques applied to the study of vacua of supersymmetric theories in String Theory-engineered models.
- Conducted original research on graph-encoded data, developing novel deterministic computational tools.

Imperial College London

Master of Science in Physics

Oct 2021 – Oct 2022

London – UK

- Graduated with *Distinction* (1st of my cohort).
- **Relevant Coursework:** Programming Skills (Matlab, Labview, Mathematica).
- **Prizes:** “Outstanding Performance in the MSc”, highest graduating average in my cohort (86.3%/100%).
- **Societies:** member of the Investment Society, with a Security and Education Certificate.

University of Pisa

Bachelor of Science in Physics

Sep 2018 – Jun 2021

Pisa – IT

- Graduated with 110/110 *cum Laude* (Top 10% of my cohort).
- **Relevant Coursework:** Linear Algebra, Real and Complex Analysis, Laboratory of Physics (Probability and Statistics), Computer Science.
- **Prizes:** “Medaglia del Cherubino”, for graduating with the highest grade while being in the top 10% of my cohort in every academic year.

Experience

Career Break

Jan 2026 – Sep 2026

- Career break for medical recovery and independent research in systematic trading and ML-based alpha modelling.

Goldman Sachs

Associate Risk Quantitative Strategist

Jul 2025 – Dec 2025

Paris – FR

- Developed and maintained production software supporting daily regulatory reporting and liquidity stress test metrics for synthetic asset portfolios.
- Enhanced asset-classification methodologies, improving classification accuracy by 11% across reporting datasets.
- Designed and implemented data-processing pipelines across multiple internal platforms, improving consistency and reducing manual intervention in downstream risk analytics.
- Worked with large-scale financial datasets using SecDB, Slang and SQL in a production environment.
- Collaborated with risk, technology and reporting teams to ensure robustness and reliability of regulatory calculations and reporting infrastructure.

Deutsches Elektronen SYNchrotron

Graduate Researcher

Oct 2022 – May 2025

Hamburg – DE

- Conducted original research in high-energy theoretical physics within the String Theory group, focusing on mathematically rigorous modelling of high-dimensional systems.

- Designed novel computational methods for graph-structured data arising in string theory based applications.
- Published research in peer-reviewed journals across theoretical physics and machine learning.
- Maintained and supported a Linux-based research computing environment serving multiple researchers.
- Mentored junior researchers in mathematical modelling, computational methods and scientific programming.

Research Publications

- **INSPIRE Author profile:** 2711158. Author of 6 peer-reviewed publications:
 - 5 research papers in High Energy Physics – Theory (Published in *Physics Review D*)
 - 1 paper in Computer Science – Machine Learning on Graph Neural Networks (cross-disciplinary).
- Publications have accumulated approximately 60+ citations.

Projects

Systematic Equity Trading Strategy with IBKR Execution

Python | scikit-learn | XGBoost | IBKR API

- Developed an end-to-end systematic trading framework covering data ingestion, feature engineering, model training, backtesting and execution.
- Researched predictive signals based on momentum, volatility, intraday range and liquidity-related characteristics across liquid US equities.
- Implemented walk-forward validation and rolling retraining procedures to evaluate out-of-sample performance and reduce overfitting risk.
- Trained gradient-boosted tree models for short-horizon return forecasting.
- Evaluated performance over a 10-year historical sample of daily bars including transaction-cost assumptions, achieving a backtested Sharpe ratio of 1.8.
- Built an automated execution layer using the Interactive Brokers API including position sizing, order management and risk controls.

Stock Return Prediction Challenge (2020)

Python | Pandas | Numpy | CatBoost | Random Forest

[Qube Research & Technologies](#)

- Built a cross-sectional alpha prediction pipeline for equities using an ensemble of CatBoost and Random Forest, with sector-neutralized ranking and quantile-based signal generation.
- Engineered multi-horizon financial features and implemented leakage-aware preprocessing.
- Used RFECV-based feature selection and low-correlation model blending for improved predictive performance.
- Public score of 0.5200, benchmark submission scores 0.5131.

Learning 3-Manifolds with Graph Neural Networks

Python | PyTorch | PyTorch Geometric | Deep Learning

[Deutsches Elektronen SYnchrotron](#)

- Developed a Graph Neural Network framework to classify topological equivalence (homeomorphism) of complex graph-manifolds with a resulting accuracy of 70.7%.
- Designed graph-based feature representations from JSJ decomposition (plumbing graphs) and trained GNN models to learn invariants under graph transformations.

Professional Certificates

Columbia University via Coursera

Aug 2025

[Introduction to Financial Engineering and Risk Management](#)

Coursera

- **Relevant Coursework:** Concepts in Probability and Optimization, Fixed income securities, Pricing of Swaps and Options with Binomial and Black-Scholes Models, Pricing of American Options, Futures, Forwards and Assets with dividends.

Technologies

Programming Languages: Python, Mathematica, SQL.

Machine Learning Libraries: PyTorch, PyTorch Geometric, Pandas, scikit-learn, XGBoost, CatBoost.

Tools: Jupyter, $\LaTeX$.